

CS305 Database Systems

Lecture 18: SQL

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Outline

1. Keys
2. Queries
3. Normalization

Primary Key

- A primary key uniquely identifies each row in a table.

Foreign Key

- A foreign key references the primary key of another table.

Inner Join

* An inner join returns rows matching in both tables.

Outer Join

1. A left outer join keeps unmatched rows from the left table.

Projection

2) Projection chooses which columns appear in the result.

Aggregate Function

(3) An aggregate function computes a single value over many rows.

Grouping

- GROUP BY partitions rows into groups sharing a value.

Having Clause

- HAVING filters groups after aggregation.

Subquery

- A subquery is a query nested inside another query.

View

- A view is a named query stored in the schema.

Index

- * An index is a data structure that speeds up lookups.

Functional Dependency

1. A functional dependency means one attribute determines another.

Normalization

- Normalization removes redundancy by splitting tables.

Denormalization

- Denormalization reintroduces redundancy to improve read speed.

Transaction

- * A transaction is a unit of work executed atomically.

Atomicity

1. Atomicity means all operations succeed or none do.

Durability

2) Durability means committed changes survive failure.

Isolation

- Isolation determines how concurrent transactions see each other.

Deadlock

- A deadlock occurs when transactions wait on each other forever.

Stored Procedure

- * A stored procedure is program logic stored in the database.

Trigger

- A trigger executes automatically in response to a table event.

Query Optimizer

- The optimizer chooses a low cost execution plan.

Schema

- A schema is the formal definition of a database structure.

Tuple

* A tuple is a single row of a relation.

Syntax Notes

`SELECT * FROM users WHERE id = 1;`

- Run `psql --verbose` to see the plan.
- A well-formed query ends with a semicolon.

Numeric Notes

- Temperatures of -5 degrees are stored as integers.
- The cost estimate $f(x) = 2x + 1$ grows linearly.

3.2 Cost Model

Figure 3: ER diagram

Summary

- Keys, joins, normalization, transactions

Appendix

- Additional reading is listed here.

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